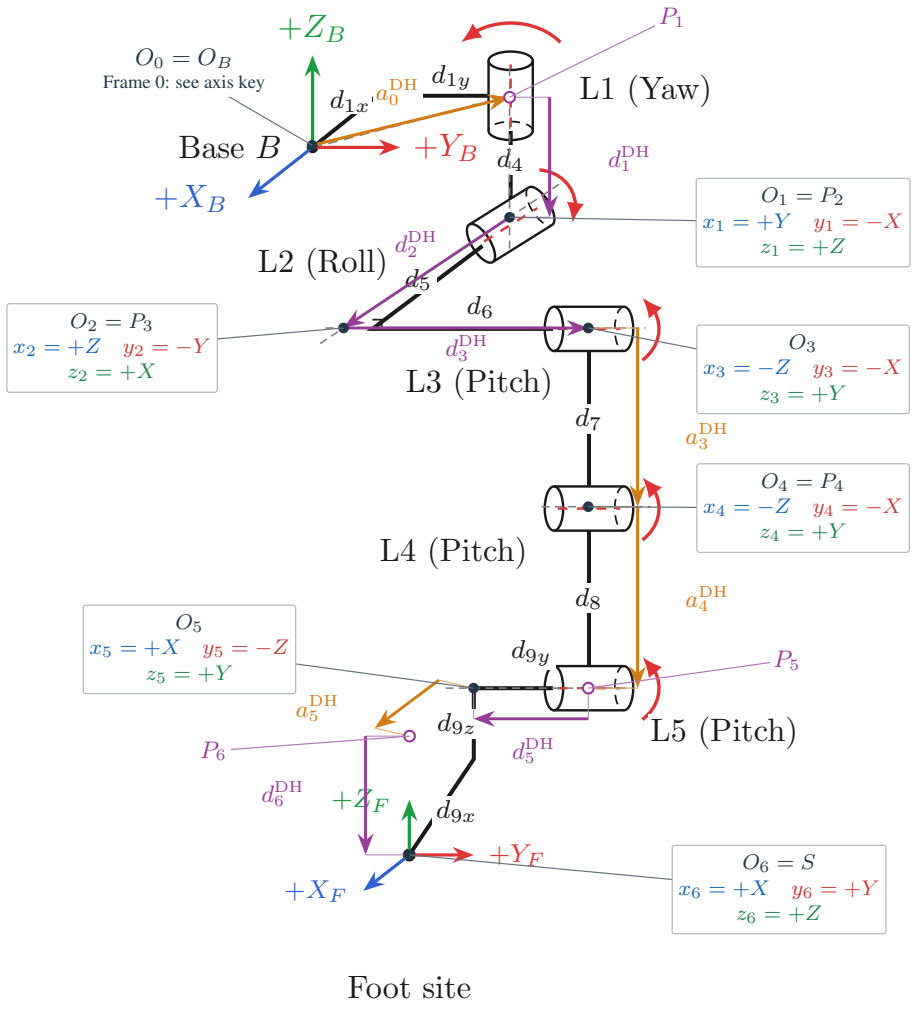
Left leg: D-H origins, construction distances and axes

Original assignment Fig. 2 retained underneath. Added labels use O_i, P_i, a_{i-1}^{DH} and d_i^{DH}; all listed lengths are in mm.



Filled dots: selected D-H origins. Open dots: intermediate P_i.
Boxed XYZ keys give directions in base B, not page directions.
Arrow paths show construction order on the original schematic.

Read each row as two translations

$$O_{i-1} \xrightarrow{a_{i-1}^{\mathrm{DH}}} \text{along } +x_{i-1} \quad P_i \xrightarrow{d_i^{\mathrm{DH}}} \text{along } +z_i \quad O_i.$$

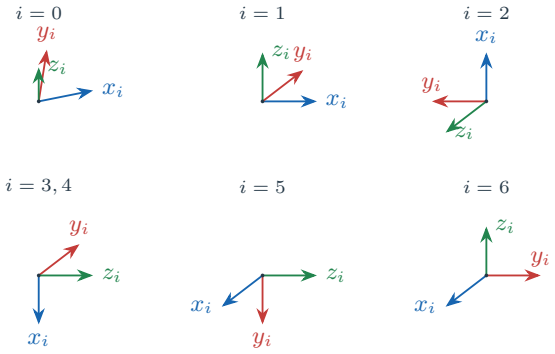
Negative values mean travel opposite to the stated positive axis.

Row	Route	a_{i-1}^{DH}	d_i^{DH}
1	$O_0 \to P_1 \to O_1$	$+39.8246$ along x_0	-0.0741 along $+Z$
2	$O_1 = P_2 \to O_2$	0	-16.060 along $+X$
3	$O_2 = P_3 \to O_3$	0	$+74.150$ along $+Y$
4	$O_3 \to P_4 = O_4$	$+78.650$ along $-Z$	0
5	$O_4 \to P_5 \to O_5$	$+78.650$ along $-Z$	-19.750 along $+Y$
6	$O_5 \to P_6 \to O_6$	$+0.500$ along $+X$	-36.225 along $+Z$

Positive coordinate axes at every origin (zero pose)

Frame	$+x_i$	$+y_i$	$+z_i$
0	$-sX + cY$	$-cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3, 4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6 (site)	$+X$	$+Y$	$+Z$

X, Y, Z denote the base axes; $s = 19/\sqrt{19^2 + 35^2}$, $c = 35/\sqrt{19^2 + 35^2}$. All frames satisfy $x_i \times y_i = z_i$.



Origin placement. O_1 is the yaw/roll axis intersection; O_2 is the roll/hip-pitch intersection. They are not the drawn cylinder centres. P_1 is a schematic yaw-axis reference at base height. For row 6, the construction takes the small a_5 step first, then d_6.
Source drawing versus signed geometry. The original is not to scale and its joint symbols do not share a calibrated 3-D projection. Gray joint-axis lines on the drawing are *unoriented*; the table and triads define the exact positive directions. Original labels $d_4, d_5, \ldots$ are handout dimension names, not D-H row numbers. In particular, $d_1^{\mathrm{DH}} = -0.0741$ is the selected origin's net height, not the CAD yaw-to-roll anchor separation.